HEADSTART: Forecasting health-system crises from published evidence when local outcome data are scarce

SNSF Ambizione application (submission: 3 November 2026) · Applicant: Dr Erol Orel · Host: Institute of Global Health, Faculty of Medicine, University of Geneva · Duration: 4 years

The problem

When a health crisis begins, the local data needed to forecast it do not yet exist. The best-performing models require years of history; they are therefore weakest exactly where decisions - arming ambulances, opening beds, triggering an escalation - are most costly and least reversible.

Yet quantitative information is available from day one: the published literature on analogous events - weather-demand associations, surge magnitudes and delays, lengths of stay and occupancy, transmission parameters. It is almost never used as formal prior information, because nobody has established whether doing so helps or harms.

The question

The project poses the question end to end: can published evidence be reliably extracted and transported to the local setting to forecast the state of the care system when local data are missing; can we detect early when it misleads; does any gain change operational decisions; and can the complete chain run automatically, verified against observed outcomes, under pre-registered rules?

The project builds on existing ground: LiteRev-Evidence, the extraction and modelling prototype under development; the data cartography built for GESICA; and the ongoing Geneva legionellosis study (BASEC 2026-00324).

Five strands

1. **Extraction** - measure the reliability of automated extraction of quantitative parameters from the literature, against a dual human-extraction benchmark, and correct the identified biases; the literature also supplies explanatory variables, recommended model families and outcome and threshold definitions.
2. **Modelling - represent the state of the care system as ordered levels of pressure, whose number and thresholds are taken first from the published evidence (by default routine / elevated / strained / critical), rather than a threshold applied to a point forecast; epidemiological models (SEIR-type) enter only for epidemiological crises, environmental crises being driven by exposure-response structures.**
3. **Evaluation** - test, historical episode by historical episode, whether literature-derived priors improve early-crisis forecasts, using at each moment only the data and the literature available at that date, with automated selection of the best model under pre-registered rules.
4. **Decision** - evaluate not statistical accuracy but decision benefit, from escalation thresholds elicited in a structured way (SHELF protocol) from dispatchers, emergency physicians and capacity managers.
5. **Automation and living review** - public data connected automatically; models re-run at each variable’s cadence; statistical assumptions checked; daily literature monitoring; at the end of the project, for validated event types and subject to authorisation, a daily observation-mode dashboard, recorded and evaluated prospectively, never decisional during the project.

How it will be tested

Three Geneva archetypes, fixed for the whole project: **respiratory epidemics** (SARS-CoV-2, influenza, RSV) carry the confirmatory test; **heatwaves** (with air pollution as co-exposure) carry the generalisation test, run only if the first succeeds; Geneva **legionellosis** is the contrasting fourth-year extension (case incidence). Two arms - with and without evidence - pass through the same pre-registered automated selection; the machinery is validated first, end to end, on **open public data** (a public benchmark), then on Geneva's operational series.

The data

Only **retrospective extracts aggregated to daily counts**: no identifiers, no individual-level data, no voice recordings, no free text. The scope is the canton of Geneva: CASU-144 dispatch data; emergency-department presentations and ICU occupancy; the cantonal physician and pharmacist and SIG for the environmental strands. Plus all other freely available data (climate, demographics, etc.). Operational access would follow the official route: a CCER submission.